Dependency-Preserving Evidence Packing for Numerical Question Answering

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Abstract

Financial questions require numbers to remain associated with their headers, units, document structure, and source links. We implement a shared answering pipeline with dataset-specific adapters and evaluate a controlled comparison of ordinary packing, dependency-preserving packing, and length-matched context expansion. The model and evidence cap are fixed within each comparison. The implementation uses prediction-only readers, deterministic lexical retrieval, validated answer plans, bounded numerical execution, and dataset-specific official scoring. Full FinQA development evaluation contains 883 questions per method; dependency packing yields two officially correct executions while ordinary packing yields none. Most original outputs fail execution validation. Training-only diagnostics motivate kind-specific schemas and compact instructions. A separately labeled single-operation arithmetic ablation is evaluated on 40 development questions each for FinQA, TAT-QA and TAT-HQA, plus 141 turns in 40 complete ConvFinQA conversations. FinQA is an authorized overlapping rerun; other expanded subsets are disjoint from prior selections. Executable coverage improves, but official scores remain near zero. MultiHiertt fails the executable training gate and HybridQA lacks required evidence. These measurements do not establish reliable packing gains.

1 Problem and motivation

A value such as "15" is difficult to interpret without its year, row label, unit, and sometimes an ancestor header. A linked passage may be relevant only through the table cell that introduced it. Conversational questions can refer to earlier answers, while hypothetical questions change the facts being used. Flattening evidence can weaken these relationships even when the correct number is present. We ask: with the same model and evidence budget, does preserving dependencies between evidence and its headers, units, hierarchy, and links improve answer quality compared with ordinary evidence packing? We do not assume improvements on every dataset. The intervention may consume space needed for other facts, and incorrect inferred headers can hurt.

2 Related work

FinQA introduces program-based numerical reasoning over financial reports (Chen et al., 2021). TAT-QA combines table and paragraph evidence with multiple answer types (Zhu et al., 2021). ConvFinQA requires sequential numerical reasoning over conversation turns (Chen et al., 2022). HybridQA tests multi-hop reasoning over Wikipedia tables and linked passages (Chen et al., 2020); it is a transfer test, not a finance-only benchmark. Learning to Imagine introduces hypothetical reasoning over financial evidence (Li et al., 2022), and MultiHiertt combines hierarchical tables and text with span/program reasoning (Zhao et al., 2022). Program of Thoughts delegates computation to an external interpreter (Chen et al., 2023). TAT-LLM uses an extractor–reasoner–executor pipeline (Zhu et al., 2024). These are established approaches. Our contribution is a controlled engineering study of deterministic dependency closure at a fixed evidence budget, not a claim that retrieval, structured evidence, or executable reasoning is new. FINDER combines generative retrieval with dynamically selected in-context examples (Khatuya et al., 2025). FinCARDS uses finance-aware schema alignment and tournament reranking to select evidence with audit traces (Zhou et al., 2026). These close approaches limit our novelty claim. We instead use deterministic lexical ranking and recursive closure of actual header, unit, hierarchy and link dependencies, with one model call and the same model/evidence cap across methods. Any benefit must survive the neighboring-context length control. The project source ledger records further related work and its verified primary links.

Dataset	Local inputs	Main challenge
FinQA	Available	Arithmetic programs
TAT-QA	Available	Spans, arithmetic, scale
ConvFinQA	Available ZIP	Predicted turn history
HybridQA	Corpus missing	Cell-to-passage links
TAT-HQA	Available	Hypothetical changes
MultiHiertt	Available	HTML hierarchy

Table 1: Input readiness. Available inputs do not imply live model validation.

Prediction-only readers → lexical retrieval → evidence packing → frozen model → validated answer plan → restricted execution → output adapter. References join only at the official evaluation stage. Earlier predicted state feeds the next conversation turn.

Figure 1: One reasoning architecture with dataset-specific I/O and evaluation.

3 Datasets and scope

All runs use the supplied document context, rather than searching a global corpus. Official splits remain separate. Readers preserve raw strings, stable identifiers, cell coordinates, original tables, paragraph offsets, and available links. TAT-HQA uses its hypothetical subset; label-derived intervention fields and related factual answers are excluded. ConvFinQA exposes only questions through the current turn and our earlier predicted state. MultiHiertt preserves HTML and merged-cell geometry and regenerates context from the HTML rather than trusting descriptions of unverified provenance. TAT-QA's released test-gold version has different IDs from its original public test. The former is evaluated using its own inputs and identifiers; position joins are forbidden. TAT-HQA's hidden test filename contains "dev", but is still treated as hidden test. FinQA and ConvFinQA share source documents; TAT-QA and TAT-HQA also overlap. We therefore retain per-dataset results and do not interpret these as six independent replications. Exact input fingerprints, source IDs, question–table hashes, and approximate near-duplicate checks are prepared without consulting answers.

4 Shared architecture

The common evidence record stores raw text, provenance, dependency IDs, unit/scale metadata, and uncertainty about inferred structure. The final model request uses a field allowlist. Answers, gold evidence, derivations, reference programs, and future turns never enter retrieval or prompts.

4.1 Retrieval and packing

BM25 lexical retrieval ranks cells and paragraphs using the current question, available earlier questions, and immediate dependency text. Ties use stable IDs. Both primary methods receive readable headers. Ordinary packing inlines immediate labels. Dependency-preserving packing recursively closes header, unit, ancestor, and link-anchor dependencies and deduplicates shared nodes. Candidates are visited in ranked order; a complete bundle is accepted only if its serialized evidence fits. Dependency cycles fail validation. Whole facts are admitted or dropped; strings are not cut. The length-matched control visits a ranked seed and then its immediate source neighbors, targeting the dependency method's actual serialized length. It uses no graph closure. Whole-fact admission can underfill the target, so we record the residual gap. It can produce a different fact mix, and is an imperfect control for length alone. Identical candidate order, counting rules, prompt, model, and decoding settings are frozen before scoring.

4.2 Answer tools and provenance

The shared plan supports extraction, multiple spans, counting distinct spans, arithmetic, aggregates, comparison, selection, and temporary hypothetical changes grounded in the question. Decimal arithmetic is bounded in magnitude, step count and exponent. Arbitrary generated code is never executed. Operands must refer to selected evidence, question spans, allowed constants, earlier steps, or selected earlier predictions. Known incompatible units are rejected; unknown units remain an explicit limitation. Predicted conversational programs are expanded locally into cumulative operations without reference intermediates. Complete selected table rows also export through the official table-aggregate operators; partial rows are rejected. Some valid local answer modes lack an official FinQA program representation; these are recorded as unsupported exports and retained as failures, without fabricating identity operations. Direct-answer diagnostics have no official program metric and use an explicitly auxiliary numeric score where necessary.

4.3 Execution infrastructure

Existing Gemini and OpenAI adapters share a durable SQLite ledger. Runs require an explicit

zero-spend gate and never switch providers automatically. Requests record exact provider/model identity, settings, raw and parsed responses, token usage, latency and status. Dispatch is committed before a network call. Completed responses are reused, including invalid predictions. Ambiguous timeouts and quota failures stop for review; transient server retries are bounded. Model drift stops paired scoring. Mock and live outputs use separate directories and labels. An earlier hosted-provider attempt stopped at token counting with HTTP 404; that diagnostic is separate from the local Ollama results reported here. The bounded local runs initially failed preflight because the required local server URL was absent from the launch environment. Restoring the recorded URL resolved that blocker without changing the model or inference contract.

5 Experimental protocol

The frozen smoke contains three predeclared FinQA development questions. A separate 20-question pilot excludes those IDs and compares all three packers with the same local model and a 3000-byte conservative evidence cap. Configurations also prepare per-dataset subsets and full splits. Conversations are sampled and evaluated as complete ordered groups, so their turn count need not equal 20. Training fixtures are engineering examples; final test inputs are separate from prompt development. The main outcome is official execution accuracy for FinQA/ConvFinQA and official F1 for the remaining datasets, with companion metrics. Missing predictions, invalid outputs, abstentions, unsupported exports and API failures stay in the denominator. Official metric functions run in isolated processes. Wrappers bypass the TAT-HQA CLI's undefined counter and retain metric logic; they preserve raw official-function scores and separately force failed outputs to zero. Numeric zero is serialized as text for TAT scoring. MultiHiertt's invalid-program flag bug is documented rather than used to obtain credit. Paired bootstrap intervals resample documents/conversations with a fixed seed. The confirmatory contrast is dependency versus ordinary packing. Neighbor expansion, direct answers, and component removals are diagnostic. Cross-dataset confidence claims must respect shared sources; no naive independent-dataset pooled interval is produced. A small pilot cannot establish broad superiority.

6 Big Data preparation

The Spark workload prepares partitioned Parquet tables for documents, questions, and facts, joins questions to documents, explodes dependency edges, checks unresolved links, and computes exact content overlap. A Python reference computes equivalent canonical results. Repeated source contexts are deduplicated, not counted as new questions. Commands prepare 100, 1,000, and full-source sizes using real records; actual counts must be reported. Measurements include wall time, sampled process-tree RSS, output size, and Spark event logs for spill/task analysis. Local Spark on one laptop is not a distributed-cluster scalability result. A verified local[1] Spark 4.2.0 run matched the Python reference: 25 joined questions, 5733 dependency edges, zero unresolved edges; Spark wall time was 8.129 seconds. The sampled process-tree peak RSS was 835276800 bytes (0.1-second sampling; shared pages may be counted twice). This is a small correctness check, not a scale or speedup result. The development overlap audit found 218 shared unique table/content keys between ConvFinQA and FinQA, and 273 between TAT-HQA and TAT-QA; labels were not used.

7 Measured results and coverage

FinQA uses the full 883-question development split. The other datasets use the original frozen development subsets: 20 questions each for TAT-QA, TAT-HQA and MultiHiertt, and 22 ordered ConvFinQA turns in complete conversations. Seed 2626 and all three packing methods are fixed; no selected examples overlap the earlier pilots. HybridQA is unevaluated: the genuine linked evidence corpus remains missing. FinQA test remains incomplete (162 saved rows, 54 per method, one unresolved inflight request) and is excluded from architectural comparisons.

Dataset	Method	N	Saved	Score
FinQA	O	883	883	0.00
FinQA	D	883	883	0.23
FinQA	L	883	883	0.00
TAT-QA	O	20	20	0.00
TAT-QA	D	20	20	0.00
TAT-QA	L	20	20	0.00
TAT-HQA	O	20	20	0.00
TAT-HQA	D	20	20	0.00
TAT-HQA	L	20	20	0.00
MultiHiertt	O	20	20	0.00
MultiHiertt	D	20	20	0.00
MultiHiertt	L	20	20	0.00
ConvFinQA	O	22	22	0.00
ConvFinQA	D	22	22	0.00
ConvFinQA	L	22	22	0.00

O: ordinary; D: dependency; L: length matched. Scores are percentages: official execution accuracy for FinQA/ConvFinQA and official F1 for the other datasets. All selected examples remain in the denominator; missing, infrastructure-failed, abstained and invalid answers receive zero. Saved coverage includes invalid answers, rather than equating response completion with valid execution.

Dataset	Method	V	A	E	I	M
FinQA	O	0	82	801	0	0
FinQA	D	4	98	781	0	0
FinQA	L	1	130	752	0	0
TAT-QA	O	0	1	19	0	0
TAT-QA	D	0	6	14	0	0
TAT-QA	L	0	3	17	0	0
TAT-HQA	O	0	5	15	0	0
TAT-HQA	D	0	9	11	0	0
TAT-HQA	L	0	10	10	0	0
MultiHiertt	O	0	10	10	0	0
MultiHiertt	D	0	15	5	0	0
MultiHiertt	L	0	13	7	0	0
ConvFinQA	O	0	6	16	0	0
ConvFinQA	D	0	7	15	0	0
ConvFinQA	L	0	9	13	0	0

V: valid executable/exportable answer; A: abstention; E: execution, plan-validation or official-export failure; I: infrastructure failure; M: missing output. A missing-only run is reported as unevaluated rather than an observed zero score. TAT-QA: status completed; saved 60/60, pending 0. TAT-HQA: status completed; saved 60/60, pending 0. MultiHiertt: status completed; saved 60/60, pending 0. ConvFinQA: status completed; saved 66/66, pending 0. finqa: dependency minus ordinary execution = 0.00227; paired document/conversation bootstrap 95% interval [0.00000, 0.00577], 299 clusters (1000 resamples; seed 2626). tatqa: dependency minus ordinary f1 = 0.00000; paired document/conversation bootstrap 95% interval [0.00000, 0.00000], 19 clusters (1000 resamples; seed 2626). tathqa: dependency minus ordinary f1 = 0.00000; paired document/conversation bootstrap 95% interval [0.00000, 0.00000], 20 clusters (1000 resamples; seed 2626). multihiertt: dependency minus ordinary f1 = 0.00000; paired document/conversation bootstrap 95% interval [0.00000, 0.00000], 20 clusters (1000 resamples; seed 2626). convfinqa: dependency minus ordinary execution = 0.00000; paired document/conversation bootstrap 95% interval [0.00000, 0.00000], 6 clusters (1000 resamples; seed 2626). Subsets and rare successes yield unstable intervals and cannot establish population improvement. Zero-width bootstrap intervals for all-zero samples reflect absent empirical variation and do not establish equivalence. Related datasets share source tables, so scores and uncertainty are kept separate. FinQA dependency packing achieved two correct executions out of 883, compared with none for ordinary and length matched; this is weak evidence rather than guaranteed improvement. No changes to prompts or outputs were tuned against evaluation answers. Failure (FinQA, ordinary, V/2008/page_17.pdf-1): Malformed structured output. Failure (TAT-QA, ordinary, 076daf33-1a7f-4255-8e27-764a43d97233): 'The Group's free cash flow grew 1.2% to S$3.65 billion.'. Failure (TAT-HQA, ordinary, 06e1818b3dfbd0d1f89818956b4c493a): Malformed structured output. Failure (MultiHiertt, ordinary, 2b1af05e4d4b4683828c1da7b0d2ab41): '23,385'. Failure (ConvFinQA, ordinary, Double_MRO/2017/page_111.pdf_0): [<class 'decimal.ConversionSyntax'>].

Local Ollama generation used qwen2.5:7b-instruct, Q4_K_M, digest `845dbda0ea48ed74` (full digest in frozen metadata), an 8192-token context, 2048 reserved output tokens, temperature zero, top-p one, seed 2626, and a 3000-byte conservative evidence cap. Prompt, schema and generation settings are unchanged for the bounded evaluation. Two independent dataset workers use separate exclusive-lock SQLite ledgers; each conversation and method history remains sequential and prediction-only. New dispatch stops after 3.5 hours, leaving at least one hour of the five-hour budget for scoring and delivery. Inflight requests are preserved and never automatically retried. Official scorer hashes and failure details accompany the measurements.

8 Recovery and output validation audit

The four original bounded runs were blocked before inference by a missing `OLLAMA_BASE_URL` environment variable. Each had zero saved or failed predictions; pending method outputs were 60 for TAT-QA, 60 for TAT-HQA, 60 for MultiHiertt, and 66 for ConvFinQA. No inference ledger existed for those runs. The recorded local Windows endpoint was restored and its model digest verified; a real TAT-QA request completed before bulk dispatch. That exact request is reused from its ledger, including its original response and attempt metadata. The frozen IDs, ordering, prompts, evidence cap, schema and decoding settings were retained. No FinQA generation was repeated.

The saved-output audit examined all 1344

FinQA development rows labelled "Malformed structured output", plus 92 partial-test rows, 35 pilot rows and 8 smoke rows. All were valid JSON satisfying the broad response schema, but violated the kind-specific plan contract. For example, table aggregation omitted its required row, and extraction returned arithmetic steps instead of its required operand. Every malformed row failed these declared plan rules; none passed the rules and was then rejected. The schema admits combinations that the interpreter disallows; this is a limitation of generation constraints, not evidence that these returned plans have valid executable meanings. Relaxing the executor or deleting fields would reinterpret model answers. Accordingly the original FinQA scores remain unchanged, and no corrected inference rerun is mixed with them. The generic error label obscures this distinction; the detailed audit is saved separately.

8.1 Revised evaluation and root-cause diagnosis

Training diagnostics exposed a mismatch between the broad output schema and the executable contract. Kind-specific schemas improved structural guidance. Versions 4–5 restricted arithmetic to a single operation; this is a capacity-limited ablation, not a general reasoning model. The final v5 measurements are summarized below. All original and intermediate results remain in the archived artifacts.

Dataset	Method	N	Exec	Correct	Score
FinQA	O	40	10	0	0.00
FinQA	D	40	7	0	0.00
FinQA	L	40	10	0	0.00
TAT-QA	O	40	11	0	0.45
TAT-QA	D	40	5	0	0.45
TAT-QA	L	40	11	0	0.00
TAT-HQA	O	40	14	0	0.00
TAT-HQA	D	40	14	0	0.00
TAT-HQA	L	40	16	0	0.00
ConvFinQA	O	141	38	0	0.00
ConvFinQA	D	141	21	0	0.00
ConvFinQA	L	141	29	0	0.00

Score is official execution accuracy for FinQA/ConvFinQA and official F1 for TAT-QA/TAT-HQA, in percent. Exec counts local executions, including outputs without a supported official program export. Correct counts official execution matches or exact matches. Methods O/D/L denote ordinary, dependency and length-matched packing. These are development diagnostics: FinQA reuses an overlapping 40-question subset of the original full dev run; the other subsets exclude earlier frozen selections. ConvFinQA contains 141 turns in 40 complete conversations. Missing and failed outputs remain in the denominator.

MultiHiertt v5 produced no executable training plans and was not scaled to development evaluation. HybridQA remains blocked by missing linked evidence. Neither is assigned a fabricated development score.

Why the scores collapse. Across the 783 saved v5 outputs, the following counts are derived directly from execution status and error messages. These categories describe interpreter failures, not semantic accuracy.

Dataset	Nonnum.	Partial	Placeholder	Exec
FinQA	43	25	10	27
TAT-QA	21	7	14	27
TAT-HQA	28	14	19	44
ConvFinQA	107	27	76	124

Nonnum. means the selected source substring could not be parsed as a number; Partial means only part of a numeric literal was selected; Placeholder means the model copied the instruction text "exact supplied id instead of a supplied identifier. Additional failures include invalid spans, unknown evidence IDs, invalid table-row selections and abstentions. Some executable outputs calculate unrelated constants. Thus execution coverage alone does not demonstrate answer correctness. The one-operation restriction also prevents representing many growth, average and hypothetical calculations.

Implemented recovery, pending live measurement. A separate opt-in quote-grounded mode replaces long evidence IDs with local IDs and character offsets with exact source quotes. Quotes must occur exactly once in the selected fact; the code resolves their positions before the existing interpreter validates and executes them. Up to 16 operations and prior-step references are supported. Invented quotes, ambiguous matches and partial numeric operands are rejected. Existing modes and scores are preserved. A bounded recovery runner compares this mode with a direct-answer diagnostic on frozen development inputs. FinQA direct numeric checks remain auxiliary and are not official program scores. No new accuracy improvement is claimed: local inference was unreachable during this repair session.

8.2 Measured recovery development diagnostics

Separate recovery run; direct FinQA numeric accuracy is auxiliary, grounded FinQA accuracy uses

the official program executor. Development samples are small and FinQA overlaps original dev. All selected questions, including missing outputs, remain in the denominator.

Dataset	Mode	Method	N	Score
finqa	direct	O	10	10.00
finqa	grounded	D	10	0.00
finqa	grounded	O	10	0.00
tatqa	direct	O	10	10.00
tatqa	grounded	D	10	0.00
tatqa	grounded	O	10	0.00
multihiertt	direct	O	10	10.00
multihiertt	grounded	D	10	10.00
multihiertt	grounded	O	10	10.00

9 Reproducibility

The project includes configuration-driven freeze, preparation, mock, live, evaluation, overlap, resource-measurement and aggregation commands. Frozen files contain IDs, seeds, source hashes, input hashes, prompts, schemas, decoding settings and code hashes. Changed code or data requires a new versioned freeze. Provider metadata and resolved models are pinned across resumes. Evaluator sources and dependencies are hashed. The environment snapshot and exact ordinary-WSL sequence are saved with the project. Credentials are excluded from delivery artifacts and model inputs. The original implementation and supplied LaTeX files are preserved; this report copies the lowercase style filename for case-sensitive compilation.

10 Limitations

Most saved model outputs are invalid or abstentions. The HybridQA linked evidence corpus is missing. All MultiHiertt splits now have real-input validation; its unused cached descriptions contain 61 coordinates without matching HTML cells, which remain disclosed rather than repaired with invented values. Hosted-model pretraining contamination cannot be audited. Lexical retrieval can miss paraphrases; header/unit inference is conservative and sometimes wrong. Unknown units do not prove semantic compatibility. Whole-fact packing may waste budget, provider counts include different framing, and context matching is approximate. Development subsets are small and cannot establish full benchmark performance. Hidden-test scoring availability remains unverified; only development results are reported. No training, distributed cluster, six-dataset validation, or novel reasoning architecture is claimed. The restricted arithmetic ablation cannot represent general multi-step calculations. Its effects combine instruction, schema and supported-plan restrictions; they do not isolate a single causal component. Local execution does not guarantee official exportability or answer correctness.

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